
Person-blind signal dominates forced-choice evaluation of interview-grounded twins

Project DOPPLER (registered on OSF as TWOPPLER)

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Methods paper — DRAFT for owner review, 2026-07-28.

Every number below links to the report or script that produced it. Nothing here is a new analysis; this is assembly of a record that is already committed.

1. The point, up front

We tried to measure how well a language-model “twin” of a real person predicts what that person says in a later interview. The measurement instrument was forced choice: show the twin a real question from a held-out interview, show it the person’s real answer hidden among three wrong ones, and see if it picks the right one.

It does not measure the twin. We built four versions of that instrument, each fixing the previous version’s flaw, and in all four a *person-blind* baseline — a prompt with no name, no excerpts, no date, no information about the person at all — solved essentially every item.

round	where the wrong options came from	person-blind accuracy	what gave it away
1	other people’s answers to other questions	17 / 17	topical coherence
2	the same subject’s answers to other questions	10 / 10	responsiveness to the question shown
3	generated answers to the <i>same</i> question	15 / 15	register, world-truth, deixis
4	round 3 + four targeted fixes, subjective items only	8 / 8	register <i>inverted</i> , entity deployment, stance-vs-premise fit

Chance is 0.25 in every row. Sources, in order: round 1 · round 2 · round 3 · round 4.

Round 4 ran under a kill rule the owner committed to in writing before the round launched. It fired. There was no round 5.

The claim we make from this is narrow and we hold it to that width: **on this corpus, with these four constructions, forced choice over a person’s verbatim answers is dominated by signal that requires no knowledge of the person.** We do not claim forced choice is dead as an evaluation format everywhere.

The paper also reports what replaced it — open-ended generation scored on two independent channels — together with the full validation trail of that replacement, including the parts that failed.

2. What the task is, and what the field does

A **twin** here is a language model whose prompt contains records of one specific real person, instructed to answer as that person (PREREGISTRATION.md §2). **Fidelity** is agreement with the person's real answers on material the twin never saw. **Lift** is fidelity minus the fidelity of a zero-information baseline — the same model, the same items, no grounding, identity redacted. Lift is the only headline metric in this project; a raw fidelity number is never reported without its baseline.

The design target is Park et al. (2024), “Generative Agent Simulations of 1,000 People”, commercialised by Simile: an agent grounded in a two-hour interview reproduces a person's survey answers at roughly 0.85 of that person's own two-week test-retest consistency. That work uses surveys. We wanted the same question asked on natural speech, so we used public interviews, where the held-out data already exists — ground a twin on a person's earlier interviews, test it on real questions from a later one.

The pre-registered instrument was forced choice (PREREGISTRATION.md §3, Stage 2): extract question–answer pairs from the held-out interview, hide the real answer among three or four distractors drawn from other guests' answers to similar questions, randomise position, and score whether the twin picks it. It was chosen because it is cheap, automatic, and has an obvious chance rate. Amendment 1 added distractor controls — matching, an entity-stripped re-score, an adversarial filter. That is the instrument the four rounds below tested to destruction.

3. Corpus and subjects

MediaSum: 463,596 NPR and CNN interview transcripts (CNN 414k, NPR 49k), 2000 to 2020 — corpus recon, with parsing rules and checksums in the index and a 20-guest hand audit in the quality note. Curation yields **578 clean candidate subjects** with at least three deduplicated substantive interviews and at least 180 days of span, of which **137 are confirmed long-tail** — no Wikipedia article under any spelling (curation report).

All four rounds ran on **development subjects only**: six subjects drawn by a seeded rule (seed 47, 578-row pool), three with a Wikipedia page and two long-tail, plus one added after a burn event (round 1 §1). One of the six yields no usable question–answer pairs and is excluded from every prediction set. Dev subjects are excluded from every confirmatory analysis permanently (Amendment 2 B4.1). No confirmatory subject was touched by any round reported here.

Item counts are small by design — 17, 10, 15 and 8. Enough to see a ceiling, not enough to size an effect, and every round report says so in its own header.

4. What the instrument claims to measure

From round 3 onward the options are generated, not harvested, so the claim being scored changes and the change is binding on every write-up (Amendment 2 B10.2):

The claim scored is that the twin **identifies the person's actual position among plausible alternative positions on the same question** — not that it picks a verbatim transcript answer.

That reframing matters for reading rounds 3 and 4. A twin that reproduces a person's stance but not their words should score well. The problem the rounds document is not that the instrument is too strict about wording; it is that the real answer is identifiable as *real* without any reference to who said it.

5. The four rounds

5.1 Round 1 – distractors from other people

Three wrong options per item, drawn from other guests' answers to unrelated questions, similarity-ranked. 17 items over five subjects.

Both arms, side by side (round 1 §5, driver experiments/stage2_pilot.py):

arm	items scored	accuracy	probability mass on the true option
twin, identity redacted	17	1.000	0.972
twin, named	17	1.000	0.985
zero-information, redacted	17	1.000	0.983
zero-information, named	17	1.000	0.991
imposter twin (another person's transcripts)	17	0.941	0.885

Twin minus zero-information is **0.0000** in accuracy, by construction: both arms sit on the ceiling. Twin minus imposter is **+0.0500** accuracy and **+0.0853** probability mass, which is the only structure anywhere in the table and is not evidence about fidelity either, because the imposter arm is simply the one arm not pinned at 1.00.

Amendment 1's adversarial filter — drop every item the zero-information arm got right — removed **all 17 items**. The empty table is the filter working correctly and saying the instrument is unusable.

The mechanism is visible in the scorer's own words: *"Option B directly addresses the host's question about American responsibility."* It is matching topic. The median distractor's question-similarity to the real question was cosine 0.050, and one item of eighteen had any distractor above 0.10.

What round 1 established: the pipeline runs end to end — draw, chronological split, question-answer extraction, distractors, imposter donors, five-arm rendering, both leakage guards, cluster execution, ingest, scoring. **What it established about twins:** nothing.

5.2 Round 2 – distractors from the same person

One change: every wrong option is a real answer **the same subject** gave in one of their other interviews. Speaker is now controlled by construction.

The zero-information arm still solved every item (round 2 §1):

```
candidate items built          10
zero-information accuracy, frozen parser 1.00 (8 of 8 parsed)
zero-information accuracy, honest reading 1.00 (10 of 10, 2 recovered)
mean probability mass on the true option 0.94
smallest margin over the best rival +0.80
items rejected by the build-time gate 8 (10 on the honest reading)
items surviving into a final item set 0
```

No twin, imposter or contamination number exists for round 2. The prediction phase was never run — the stop condition was "five or more of ten rejected", and eight were. The final item set would have been empty regardless.

The diagnostic decomposition is the part that mattered. Three conditions, same ten items, direction only, not effect size (round 2 §3, implementation src/doppler/diagnostics_v2.py):

condition	question shown?	options	accuracy	mean p(true)	mean margin
standard	yes	standard	1.00	0.94	+0.91
zero-information					
entity-stripped	yes	stripped	1.00	0.87	+0.82
(names, numbers, dates removed)					

condition	question shown?	options	accuracy	mean p(true)	mean margin
question-blind (host question removed)	no	standard	0.10	0.24	-0.12

Read it as: **named entities carry none of the solve; the whole thing rides on the true answer being the only option that is responsive to the question shown.** Strip every name, number and date and accuracy does not move. Remove the question and accuracy falls below the 0.25 chance line, with the margin going negative and the model hedging.

That is intrinsic to the shape “forced choice among verbatim real answers”. Distractors are, by definition, answers to other questions. No further distractor-sourcing rule repairs it, and this evidence is what forced the instrument change registered as Amendment 2 B10.

Two secondary constraints surfaced here and both stayed true afterwards. First, the anti-leak rule starves the item pool: 32 of 100 harvested answers were excluded because the twin’s own grounding block could quote them. Second, a similarity floor cannot rescue the design at this scale — a floor of 0.10 admits **zero** items from these pools, 0.05 admits three.

5.3 Round 3 – generated counterfactuals to the same question

Every option is now a *written* answer to the same question, taking a position that genuinely conflicts with the subject’s real one. Every option, the real one included, passes one identical neutralising paraphrase, so corpus style cannot mark the true option. Distractors are checked for genuine conflict. The generator (gemini-3.5-flash-lite) never scores anything.

Responsiveness is gone by construction. The ceiling did not move (round 3 §1):

```

items built                                15    (of 17 eligible)
zero-information accuracy, frozen parser  1.00    (3 of 3 parsed)
zero-information accuracy, honest reading 1.00    (15 of 15, 12 recovered)
mean probability mass on the true option   0.77
mean margin over the best rival            +0.69   (range +0.30 .. +0.96)
items surviving into a final item set      0

```

Again no prediction phase, so no twin or imposter numbers exist for round 3.

Three replacement tells, each quoted from the scorer’s own reasoning:

- **Register.** “Options A, B, and C are written in a highly argumentative, polemical style... Option D, however, reflects the typical speaking style of a social science professor in a televised interview.” Asked for a genuinely conflicting position, the generator writes a confident thesis. Real interviewees qualify, digress, and say “it’s mixed”. The paraphrase cannot fix this, because it must preserve every substantive claim and the confidence lives in the claims.
- **World-truth.** “Option C reflects the actual political reality of that moment... Options A, B, and D suggest a competitive race, which does not align with the historical consensus.” If the person’s real position is simply correct about the world, then every genuinely conflicting alternative is wrong about the world, and a well-informed scorer picks the true one on general knowledge without modelling anyone. This is structural, not a bug in one item.
- **Deixis.** The tightest item in the set (+0.30) was solved partly on the paraphrased true answer retaining the host’s first name — a leak of conversational packaging that only the real answer has, because only the real answer was spoken to a person.

Two hypotheses about *other* tells were frozen before the gate completions were read, and are reported whichever way they landed. The **polar-question** hypothesis — that on a yes/no question all three distractors must land on the same side, making the true answer the lone dissenter — was **rejected**: polar items 7/7 solved at mean margin +0.633, non-polar 8/8 at +0.740, so the polar items were solved with *smaller* margins than the others, the opposite of the prediction. The **naming-register** hypothesis was **not supported as a systematic property**: the capitalised-entity-overlap proxy was positive on only 4 of 15 items — but the

proxy was case-sensitive and missed the case that prompted the flag, so the honest reading is that register marks the true option through conversational deixis, not through entity naming.

5.4 Round 4 – every fix at once, under a pre-committed kill rule

Round 4 attacked four of the six tells a frontier-model rater had named on round 3’s material, under the design frozen in SPEC v1.10, D6-v4:

- generation few-shot conditioned on **the subject’s own real answers**, to stop the generator writing advocacy;
- **deixis stripped** from all four options or from none, never from some;
- an explicit **plausibility check** on every surviving distractor (PLAUSIBLE / FALSE / FRINGE, only PLAUSIBLE accepted), to blunt world-truth;
- **subjective items only**, since round 3’s widest margins sat on factual questions.

Result, under both parser readings (round 4 §1):

reading	prompts	parsed	zero-information accuracy	mean p(true)	mean margin
frozen parser (the contract)	8	2	1.00	0.750	+0.630
widened reading (reported beside, never instead)	8	8	1.00	0.694	+0.558

Readings disagreeing on any item: **0**. Items surviving into a final set: **0**. No twin, imposter or contamination number exists for round 4.

The fixes worked and it did not matter. Mean probability mass fell from 0.776 to 0.694 and the mean margin from +0.690 to +0.558. Every item was still solved. An instrument that only gets harder to solve *confidently* is not getting harder to solve.

Stated so it is not over-read: round 4 is **not** apples-to-apples with round 3. Round 4 is eight subjective items; round 3 was fifteen mixed ones, and round 3’s widest item (+0.96, the world-truth case) is excluded from round 4 by design. The *direction* — 1.00 either way — is comparable. The *size* of the margin drop is partly a change of item mix.

The register fix inverted the tell instead of removing it. Conditioned on the subject’s own speech, the generator stopped writing op-eds and started writing briefing documents. The real answer says the U.S. failure to act gave Russia, Assad and Iran “an absolute free ride”; the generated alternatives reach for “the intelligence community in Langley and the Pentagon”, “recent UN Security Council resolutions”, “the Joint Chiefs of Staff”. The scorer picked the real one on “*the biting irony*”. Entity counts were matched within ± 2 and six of eight items sat at the tightest matching rung — matching on a count is not matching on how entities are *deployed*.

6. The tell taxonomy

Six mechanisms, none of which requires knowing the person. Each row names the round that exposed it and what the next round’s fix did to it.

tell	what it is	exposed by	fix attempted	outcome
Topical coherence	the real answer is the only option about this subject matter	round 1	draw distractors from the same person	removed as the dominant tell; topic still separated where a career spans many subjects (round 2 §2.1)

tell	what it is	exposed by	fix attempted	outcome
Responsiveness	the real answer is the only option that answers <i>this</i> question	round 2 — question-blind accuracy 0.10 vs 1.00	generate all options as answers to the same question	removed by construction
Register — advocacy vs hedging	generated positions sound like op-eds; real interviewees hedge	round 3 §2.1	few-shot on the subject's own answers	inverted — options became bland institutional prose and the real answer was still the one that sounded like a person (round 4 §2)
World-truth	when the real position is correct about the world, every conflicting option is wrong about it	round 3 §2.2	plausibility check + subjective-only items	2 rejections fired (1 FALSE, 1 FRINGE); not decisive in round 4, but the item most exposed to it was excluded by design, so the tell is untested rather than beaten
Deixis	the real answer carries the host's name and conversational address, because it was spoken to a person	round 3 §2.3	strip host names and address from all four options or none	removed — stripped on 8 of 8 items, no round-4 trace cites it
Real-voice idiom and entity deployment	a person with a view uses idiom and names the actors they are actually talking about; generated text name-drops institutions	round 4 §2	none — this is what the register fix produced	unfixed

A seventh, **stance-vs-premise fit**, sits across rounds 3 and 4: a leading question invites a stance, and the rule requiring every distractor to conflict with the real position puts all three distractors on the side the host did not invite. Restricting to subjective items makes this worse, not better — leading questions are more common there.

An eighth, **twin-pair stance inference** (reasoning across two versions of the same question inside one rating sheet), was named by the rater and closed by a standing rule: no rater and no scorer ever sees both twins of a duplicated question, asserted at export rather than assumed (SPEC v1.10 D6-v4.9; verified in `results/stage2_pilot4/twin_check.json`).

The taxonomy has a shape. Rounds 1 and 2 fail because a real answer is recognisable as *the answer to this question*. Rounds 3 and 4 fail because a real answer is recognisable as *something a real person actually said*. Removing one mechanism reveals the next.

7. The kill rule, and how it was applied

The rule was recorded by the owner on 2026-07-27, **before round 4 launched and before any round-4 data existed** (SPEC v1.10 D6-v4.6, quoted verbatim in round 4 §0):

KILL RULE, pre-committed before any round-4 data existed: if round 4's zero-information argmax accuracy is ≥ 0.90 , four-way forced choice is **DEAD** on this corpus and there is **no round 5 on any axis**. Rounds 1, 2 and 3 solved 17/17, 10/10 and 15/15 by three different mechanisms; a fourth instrument that also fails is evidence about the format, not about the next patch.

Round 4 measured 1.00 under both parser readings. The rule was read on the frozen number, as written. Phase 2 was not submitted. There is no round 5.

Two supporting facts belong here rather than in a footnote.

A gate-loosening option was on the table and was not taken. A margin-relaxed gate — reject an item only when the person-blind arm solves it by more than some margin — was specified in advance and explicitly *not adopted for round 4*, on the grounds that a kill rule means nothing if the bar can move in the round that tests it. It was made available at bar-lock only if round 4 landed in the grey zone (clearly below 0.90 and clearly above 0.25). It did not: accuracy was 1.00 and the *smallest* margin in the set was +0.40, wider than round 3's tightest item. The precondition never arose.

The fallback was written before the result was known. The open-ended replacement design was committed as `results/stage2_pilot3/FALLBACK_OPENENDED_SKETCH.md` (commit 71ae352) while round 4 was still being built, and was promoted to Amendment 3 unchanged in substance after the rule fired.

8. The declared finding, and exactly how wide it is

Registered wording, from Amendment 3 C1:

Declared finding (claimable, scoped): verbatim-answer forced choice over public-interview corpora is dominated by person-blind signal — topic, responsiveness, register, world knowledge — across every distractor source tested, because a real person's answer differs from any constructed alternative in ways that do not require knowing the person. Scope: this corpus (broadcast interviews, largely expert subjects) and these constructions; stated in all write-ups with the four pilot reports as the record.

Three things sharpen it.

The leak is in the instrument, not in one scorer. A frontier model from a different family than both the generator and the scorer read round 3's option sets with no information about the speaker and identified the real option **10 times out of 10** against a 0.25 chance rate, with 9 of 10 correct rejections on controls where no real answer was present, 19/20 overall (rater line, scored and committed at ddb3ac6 **before** round 4's gate was submitted, so it is a genuine prediction). It named six tells; round 4 attacked four; two were fixed, two inverted or survived, two were untested. A twin scored against such option sets would earn credit for detecting text provenance, not for modelling anyone.

Confidence moved, accuracy did not. Mean margin fell +0.69 → +0.56 across rounds 3 and 4 while accuracy stayed at 1.00.

What the finding is not. It is not “forced choice is a bad format”. It is not a claim about other corpora, other subject populations, or constructions we did not test. Four rounds is enough to stop paying for a fifth on this corpus; it is not enough to generalise beyond it.

9. The replacement, and its validation trail

9.1 Design

Amendment 3 C2 replaces the option set with free text. The twin answers the held-out question in its own words. No options exist, so there is no option set to leak.

Scoring runs on two channels that fail differently:

- **Channel 1 — embedding similarity** between the generated answer and the person's real answer. A fixed, locally-run model, never an API model and never a scored model: `sentence-transformers/all-mpnet-base-v2`, revision `e8c3b32edf5434bc2275fc9bab85f82640a19130`, pinned in Addendum A, instrument parameter 1.
- **Channel 2 — a stance judge.** A separate model labels whether the generated answer takes the same position as the real one: SAME / DIFFERENT / UNCLEAR, under a rubric frozen by hash. Pinned as

gemini-3.5-flash, temperature 0.0, thinking_budget=0, max_output_tokens=512, rubric r2 sha256 ad050d1a...102464 (Addendum A parameters 2–3).

Two rules bind the reporting. The primary metric is **own twin minus imposter twin**, computed identically in both channels, because judge and embedding biases — verbosity, topic priors, generosity — apply to both arms and cancel in the difference. And **no headline rests on one channel alone**: a claim requires direction agreement across both, and disagreement between channels is itself the reported result.

9.2 The dev pilot and its gate – PASS, directional

The C4 validation gate required the instrument to separate own twin from imposter twin on the primary model, on dev subjects, before any bar could freeze. The readings were written before the numbers existed (PILOT_SPEC §7).

17 items, five dev subjects, five arms, both scored models (OE-1 report, driver experiments/stage2_oe1.py):

channel	model	own	imposter	own – imposter (95% CI)	subjects own > imposter	paired items
1 embedding	Gemma-4-31B-it (primary)	0.6497	0.5473	+0.1024 (0.0444, 0.1770)	5 / 5	17
2 stance	Gemma-4-31B-it (primary)	0.8125	0.7273	+0.1818 (0.0000, 0.3333)	2 / 3	11
1 embedding	gemini-3.5-flash- lite (robustness)	0.6368	0.5708	+0.0660 (0.0473, 0.1119)	5 / 5	17
2 stance	gemini-3.5-flash- lite (robustness)	0.7500	0.7143	+0.1429 (0.0000, 0.2308)	2 / 4	14

One arithmetic caveat, since the columns invite it: on channel 2 the own and imposter means are computed over each arm’s own judged items, while the difference is paired on the items where **both** sides got a non-UNCLEAR label. The difference is therefore not the subtraction of the two printed means. Channel 1 has no such gap ($0.6497 - 0.5473 = 0.1024$).

Verdict: PASS, applied mechanically against the pre-written reading — own beats imposter on the primary model in both channels, and the channels agree.

Three qualifications were published with the verdict, not after it.

1. **Directional, not powered.** 17 items over five subject clusters, one of which contributes a single item. Channel 2’s confidence interval touches zero.
2. **The UNCLEAR asymmetry is real and it thins the data.** The judge returned UNCLEAR on 6 of 17 imposter answers (0.353) against 1 of 17 for the twin (0.059) on the primary model — the imposter dodges the question in its donor’s register. That is itself an own-versus-imposter difference, but it drops the stance denominator to 11 items, and a confirmatory design must expect it. The robustness model shows the same pattern, smaller (0.177 vs 0.059).
3. **Channel 1’s separation is partly topical.** Cosine between the twin’s own grounding text and the real answer correlates $r \approx 0.74$ with the own-arm score on the pinned encoder. The design anticipated this — which is why no claim rests on channel 1 alone, and why the stance channel, which topic overlap cannot satisfy by itself, has to agree. The diagnostic itself is partial: the grounding block is ~2,000 words and the encoder truncates it, so the number describes the head of the grounding, not all of it. Two figures for that window sit on the project’s record and both are right about different things — OE-1’s caveat quotes the 512-token limit of the candidate encoders generally, while the later H7 diagnostics quote **384 tokens**, which is the pinned all-mpnet-base-v2’s configured sequence length and so is the operative truncation here (h7_diagnostics.md §2). The smaller number makes the caveat stronger, not weaker.

9.3 A judge defect, found and pinned

The first judge pass was broken and the record says so (OE-1 §9).

Running gemini-3.5-flash at 256 output tokens with no thinking setting, 82 of 85 replies came back with the explanation line cut mid-phrase. A two-budget probe found the cause: the model charges hidden thinking against the output budget and did not finish thinking at either budget — 243 of 256 tokens, then 980 of 1024, both ending in truncation.

The damaging part is not the truncation. **The label itself moved between the two budgets at temperature 0** (DIFFERENT → UNCLEAR). The v1 labels were a function of the budget, not only of the rubric.

Fix, taken before any re-run: thinking explicitly disabled (thinking_budget=0), budget 512, everything else unchanged. A determinism probe ran first — three items, two runs, 3/3 identical labels, explanations intact — and only then did the batches run. v1 and v2 agree on 72 of 85 labels (84.7%) on the same generations; v1 is retained as the defect record and used for nothing else. Both settings are now pinned parameters.

9.4 The judge trust bar – FAIL, one iteration, PASS

This is the part of the validation trail that failed, and it gets the same space as the parts that passed. Full record: AUDIT_LINES_2026-07-28.md.

The first audit was inconclusive and diagnosed as our fault. 51 rows across three sheets. Human labels on sheet A (17 rows, owner time constraint, recorded as deviation D1), an out-of-family LLM co-auditor on all 51. The two lines agreed 17/17 on sheet A. Against the judge: human 0.7647 raw / κ 0.556; co-auditor 0.7843 / κ 0.596. But the auditors had been briefed with a *paraphrase* of the task rather than the frozen rubric text — an audit-protocol defect recorded as the owner’s, not the judge’s. The owner adjudicated the four rows where the judge disagreed with a concordant auditor line: **net 2–2**, two judge-correct, two auditors-correct.

The bar was then set, in writing, before its measurement existed: the judge passes if and only if **raw agreement ≥ 0.80 AND Cohen’s κ ≥ 0.60** against a rubric-briefed auditor line on a fresh blind tranche. The rationale is on the record — the 0.76–0.78 above is a lower bound because the auditors were rubric-naive.

First measurement: FAIL. Fresh 18-row tranche (sheets D/E, seed 611, drawn only from generations unused in the first audit, key sealed). Scored by experiments/oe1_param5_score.py: **raw 0.7778 (14/18), κ 0.5789**. Both legs miss. The verdict was applied mechanically.

The four disagreements were each a failure mode the earlier adjudication had already ruled on: the judge scoring a second-order conflict instead of the question’s first-order ask (twice), reading refusal-of-premise as opposition, and matching on a side claim in a pick-one question. Three of the four were the judge over-calling DIFFERENT.

One pre-committed iteration, then re-measure on the same bar. Rubric r2 makes three targeted edits, one per diagnosed failure mode, plus a reply-format line where the judge must name the central issue it scored (and a one-line parser widening to accept it). The owner approved the diff as drafted; all three edits increase strictness. A regression rule was set before the run: **if r2 breaks more than 2 of the 14 previously-correct rows, stop — it is overfitted.**

Result (experiments/oe1_r2_score.py):

step	tranche	outcome
regression	D/E, 18 rows	0 previously-correct rows broken (rule: >2 → stop). 1 of 4 disagreements fixed. Agreement 0.7778 → 0.8333.
trust bar	F/G, 18 fresh rows, seed 613	raw 0.8889 (16/18), κ 0.7978 — PASS against the unchanged bar

Both remaining F/G disagreements are SAME-versus-UNCLEAR confusions. There is no SAME-versus-DIFFERENT flip anywhere on the tranche.

The bar never moved. It was fixed before the first measurement, missed, and met on the second attempt at the same numbers.

Three honesty notes stay attached to the PASS rather than being cleared by it:

- **Three known hard rows are still wrong.** Under r2, the judge's new central-issue line shows it *naming* the adjudicated central issue on all three residual D/E rows and still holding its call. The residual gap is the judge model's application, not the rubric text.
- **The dev supply is nearly exhausted.** The unused pool held only 4 judge-DIFFERENT rows, so the F/G tranche is 9 SAME / 4 DIFFERENT / 5 UNCLEAR by the old labels rather than balanced. Another label-balanced tranche requires fresh dev generations, not another draw.
- **Both auditor lines on the fresh tranches are LLM lines** (deviation D4, owner-directed), out-of-family from the generator, both scored models and the judge, with the frozen rubric read in full and the key never opened. They are reported as their own line and never pooled with a human line.

9.5 What the replacement has and has not shown

It has cleared a dev-scale validation gate on 17 items, and its judge has cleared a pre-committed trust bar after one documented iteration. That is all it had to do to unfreeze the bars.

It has **not** established anything about twin fidelity in this paper. The confirmatory run of the replacement instrument is a separate result with its own reports — H1 and H7 and H6 — written up in the main results paper; they are named here only so a reader does not assume the instrument went untested at scale. In brief, so the handoff is honest about outcomes as well as existence: the instrument separated own twin from imposter twin at confirmatory scale on 88 subjects in both channels and on both models, while H7 ended with the two channels disagreeing and H6 with too few eligible subjects to license any claim. The details, the misses and the magnitude bars belong to that paper, not this one.

10. Positioning: what this adds, and what it does not

Adaptive and uncertainty-guided questioning is an active research area with published results. No document in this project may claim otherwise (Amendment 2 B9.a, binding). The record we cite instead:

- **BED-LLM** (Choudhury et al., ICLR 2026) — Bayesian experimental design for adaptive LLM questioning: pick the next question by expected information gain about the thing you are trying to learn. Our Stage 1E entropy rule corresponds to their weak baseline.
- **Wang et al. (ICML 2025)** — adaptive elicitation in natural language, evaluated on OpinionQA among other tasks.
- **Su et al. (May 2026 preprint)** — adaptive interviewing for persona simulation; small effect, small scale (follow-up-grounded predictions 45.5% against 39.3% for core-only). The flag is planted and is cited.

Full references in §14.

This project claims two contributions, and this paper adds a third:

1. **A population-optimised static-script baseline** that adaptive-questioning work generally omits. Stage 1E showed it beating adaptive selection at a tenth of the compute (Stage 1E findings).
2. **Elicitation budgets priced in human time** — respondent seconds rather than item counts (time-cost note).
3. **This paper's scoped negative result about forced-choice instruments**, with the tell taxonomy and the four-round record behind it.

The project's identity follows from that list: DOPPLER is measurement and validation science for person-models — what makes a twin faithful, where it fails, how fast it goes stale — not a competing interviewer.

One scope statement carries over and is worth repeating because it is easy to overstate. Stage 1E's null for adaptive selection was measured on a closed pool of 48 fixed items with single-rating answers and no conversation. It licenses no claim about follow-up generation in open conversation, and it may not be cited as evidence that adaptive interviewing does not work (Amendment 2 B1).

11. Limitations

One corpus. Everything here is MediaSum: American broadcast news interviews, 2000–2020, guests who are largely subject-matter experts speaking in a professional register. The tells we found are plausibly corpus-specific. A corpus of ordinary people speaking casually might behave differently, and we did not test one.

Public personas, not private people. Stage 2’s honest object is what a person says in a public interview given their previous ones. That sentence belongs in every write-up of this project and it belongs here.

The human detectability line was waived. Amendment 2 B10.8 required the owner to attempt to spot the real answer among generated ones on 20 unmarked items. The owner declined on 2026-07-27 and substituted an out-of-family LLM rater. **No human hit rate exists and none is fabricated.** This is a documented deviation, recorded in round 4 §3.1 and in Addendum A precondition 4. The LLM line answers a related but different question — whether a capable model can spot the real answer — which predicts scorer behaviour more directly but is not the check the amendment asked for.

Model-family relationships, declared (Amendment 2 B10.3, carried into Amendment 3 C3). The option generator in rounds 3 and 4 was `gemini-3.5-flash-lite`, which is also the registered robustness scorer — an owner cost decision, declared at the time. It was inert in those rounds, since only Gemma scored anything there, but it would be live at confirmatory scale. Under the open-ended instrument the stance judge (`gemini-3.5-flash`) is a different version from the robustness scorer (`gemini-3.5-flash-lite`) but shares its family. The consequence is registered: **robustness-arm absolute scores are explicitly secondary; only the own-minus-imposter contrast carries robustness weight.**

Dev scale throughout. 17, 10, 15 and 8 items on five or six development subjects. These are directions and ceilings, not effect sizes. The ceiling finding survives the small numbers because a ceiling at 1.00 with margins from +0.40 to +1.00 is not a marginal call — but nothing else in these rounds should be read quantitatively.

A measurement artifact we chose not to patch mid-flight. The scoring model sometimes prints its probability distribution twice, which the frozen parser rejects as malformed. The rate climbed with option length: 2 of 170 in round 1, 2 of 10 in round 2, 12 of 15 in round 3, **6 of 8 in round 4**. Every affected reply was recoverable and every recovered reply agreed with the frozen reading, so no conclusion here turns on it — but in round 4 the contract number rested on two of eight replies, and a run where the two readings disagreed would be a serious problem. Widening the parser was treated as a bar-lock decision, not an implementer’s; the instrument died before it was taken.

One unaudited data-quality risk. Round 2’s answer pool was widened to include non-substantive transcripts (mostly older CNN panel shows) to keep the item yield above zero. That is exactly the material where MediaSum is most likely to misattribute speech to the wrong speaker, which would silently break the one property the same-subject design exists to guarantee. It was flagged and not audited (round 2 §4).

“Pre-registered” means two different things here, and the registration is retrospective for this paper. The governance documents are frozen in git with per-document commit and sha256 provenance, and the OSF registration is now live (§13). But the registration was made on **2026-07-28**, after every round reported in this paper had run. For the four forced-choice rounds and the open-ended dev pilot, “pre-registered” therefore means **“committed to version control before the data was touched”** — the per-document git commits and sha256es in snapshot v4 are the before-data evidence, and the OSF deposit is an external timestamp added afterwards. That is weaker than a registration made in advance, and it is stated rather than blurred.

12. Cost

The whole programme reported in this paper, from `results/cost_log.jsonl`, one line per run:

item	compute (node-hours)	API spend	calls the spend covers	all ledger calls
Round 1	0.3544	\$0.00	639 scored cluster calls	661
Round 2	0.2633	\$0.00	30 scored cluster calls	30
Round 3	0.1156	\$0.0999	239 priced API calls	305
Round 4	0.1064	\$0.0447	137 priced API calls	145
Open-ended dev pilot (OE-1)	0.1053	\$0.3177	340 priced API calls	425
Judge r2 round (re-judge + re-score)	—	\$0.0459	36 priced API calls	36
total	0.945	\$0.508		1,602

The two call columns differ on purpose, and here is exactly how. The fourth column counts only the calls the dollar figure beside it pays for, which is why every dollar figure matches the ledger to the cent. The fifth column is the raw sum of `n_calls` for that run prefix in `results/cost_log.jsonl`. Four rows differ:

- **Round 1** — 639 is prediction (170) plus classifier (469). The ledger’s 661 adds a **22-call smoke slice** that produced no scientific output. Note the asymmetry in that row: the smoke slice carries 0.2280 of the 0.3544 node-hours, so the compute column *does* include it while the scored-call column does not.
- **Round 3** — 239 is the priced flash-lite work (build 175, budget probe 15, controls 49). The ledger’s 305 adds the **51-call abandoned Pro generator line** (superseded, unpriced) and the **15 cluster gate calls** (no API cost).
- **Round 4** — 137 is the priced flash-lite build. The ledger’s 145 adds the **8 cluster gate calls** (no API cost).
- **OE-1** — 340 is the four priced API runs (flash-lite generation, and three judge passes at 85 calls each). The ledger’s 425 adds the **85 cluster generation calls** for the primary model, which are billed in node-hours, not dollars.

Round 2 and the r2 round have no such gap; their two columns agree.

Three notes the ledger insists on. GPU time is billed from the scheduler’s own accounting, never from the in-process wall clock — a node is billed whole from allocation, so a failed attempt costs what a successful one of the same length costs (round 2 wasted 0.0558 node-hours to a node fault, and it is billed). One abandoned generator line carries a **null** cost because the price table has no entry for that model: **a missing price, not a zero cost**. Real money was spent on it, and that is why 51 of round 3’s calls sit outside the priced column rather than inside it at zero. And the totals row is a sum of measured lines, not a budget: 0.3544 + 0.2633 + 0.1156 + 0.1064 + 0.1053 node-hours, and \$0.09988 + \$0.044708 + \$0.317696 + \$0.045888 across the priced runs.

Killing an instrument after four rounds cost under a node-hour and under a dollar. The expensive resource in this project is owner review time.

13. Where every number comes from

what	document	regenerated by
Round 1 — other-people distractors	stage2_pilot/PILOT_REPORT.md	experiments/stage2_pilot.py
Round 2 — same-subject distractors, decomposition	stage2_pilot2/PILOT_REPORT_2.md	experiments/stage2_pilot2.py
Round 3 — generated counterfactuals	stage2_pilot3/PILOT_REPORT_3.md	experiments/stage2_pilot3.py
Round 4 — fixes + kill rule	stage2_pilot4/PILOT_REPORT_4.md	experiments/stage2_pilot4.py
Round 4 design contract	stage2_pilot4/SPEC_v1.10.md	frozen snapshot
Frontier-rater detectability line (pre-gate)	stage2_pilot4/DETECTABILITY_RATER_LINE.md	—
Open-ended dev pilot (OE-1)	stage2_openended/OE1_PILLOT_REPORT.md	experiments/stage2_oe1.py

what	document	regenerated by
Judge audit, trust bar, FAIL → PASS	stage2_openended/AUDIT_LINES_2026-07-28.md	oe1_param5_score.py, oe1_r2_judge.py, oe1_r2_score.py
Judge rubric r2 (pinned)	stage2_openended/rubric_r2_draft.txt	frozen, sha256 ad050d1a...102464
Corpus recon and curation	stage2_corpus_recon.md, stage2_curation_report.md	experiments/mediasum_index.py
The contract: stages, hypotheses, bars	PREREGISTRATION.md	frozen
Instrument revision (B10), positioning (B9)	PREREGISTRATION_AMENDMENT_2.md	frozen
Kill record (C1), replacement (C2), validation gate (C4)	PREREGISTRATION_AMENDMENT_3.md	frozen
Frozen instrument parameters and judge trust bar	PREREGISTRATION_AMENDMENT_2_ADDENDUM_A.md	frozen
Cost ledger	cost_log.jsonl	every run driver
Governance snapshot deposited for timestamping	osf_preregistration_snapshot_v4.md	frozen
Errata against the frozen documents	PREREGISTRATION_ERRATA.md	appended, never edited

Pre-registration deposit. Registered **2026-07-28** at <https://osf.io/qz28m>, on the associated project <https://osf.io/74bq3>. The registration carries the name **TWOPPLER**; **DOPPLER** is the internal codename used throughout the pre-registration, the results record and the `src/doppler` package. They are the same project.

What the deposit covers, stated from the repository record. The registration **postdates** Stage 1, Stage 1E and the Stage 2 H1/H7 confirmatory run — for those it is **retrospective**, and the before-data evidence remains snapshot v4’s per-document git commits and sha256es. It **predates** the H6 confirmatory-subject scoring, the H5 substituted analysis and the $D_{\min} = 3$ arm — for those it is **prospective**. Everything in *this* paper falls in the first group.

[registration summary, verbatim: pending — the registration is inside OSF’s approval window and not yet publicly readable; to be pasted by the owner]

14. References

The three works named in Amendment 2 B9.a, resolved to full citations. Each was confirmed against the arXiv listing page for the identifier given; the amendment names them only by short description.

- Choudhury, D., Williamson, S., Goliński, A., Miao, N., Bickford Smith, F., Kirchhof, M., Zhang, Y., & Rainforth, T. (2026). *BED-LLM: Intelligent Information Gathering with LLMs and Bayesian Experimental Design*. International Conference on Learning Representations (ICLR) 2026. arXiv:2508.21184. <https://arxiv.org/abs/2508.21184>
- Wang, J., Zollo, T., Zemel, R., & Namkoong, H. (2025). *Adaptive Elicitation of Latent Information Using Natural Language*. International Conference on Machine Learning (ICML) 2025. arXiv:2504.04204. <https://arxiv.org/abs/2504.04204> — evaluated on the Twenty Questions game, adaptive student assessment, and dynamic opinion polling on **OpinionQA** (Santurkar et al., 2023), which is the evaluation B9 refers to.
- Su, R., Liu, Y., & Hu, J. (2026). *Adaptive Interviewing for Persona Simulation in LLMs: Evidence-Grounded Reasoning Improves Decision Alignment*. Preprint, submitted 28 May 2026. arXiv:2605.29458. <https://arxiv.org/abs/2605.29458> — not peer-reviewed at time of writing.

Two further works are cited in the text and are not part of B9’s list:

- Park, J. S., Zou, C. Q., Shaw, A., Hill, B. M., Cai, C., Morris, M. R., Willer, R., Liang, P., & Bernstein, M. S. (2024). *Generative Agent Simulations of 1,000 People*. arXiv:2411.10109. <https://arxiv.org/abs/2411.10109> — the design target named in PREREGISTRATION.md §1, commercialised by Simile. Verified against the arXiv listing 2026-07-28.

- Santurkar, S., Durmus, E., Ladhak, F., Lee, C., Liang, P., & Hashimoto, T. (2023). *Whose Opinions Do Language Models Reflect?* ICML 2023. arXiv:2303.17548. <https://arxiv.org/abs/2303.17548> — the origin of OpinionQA, already on the project’s record in `results/lit_check.md`.

Draft. Not submitted, not published. All pilot numbers are development-subject measurements; no confirmatory subject was touched by any round reported here.